Pei Zhang

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Education

2019/09 - 2025/05 **Ph.D. in Mathematical Statistics, University of Maryland, College Park** (UMD)
Dissertation title: *Mixed Modeling Approaches for Characterizing Genetic Effects and Heritability Metrics in Longitudinal Phenotypes* (NIH Outstanding Graduate Research)
Primary Advisors: *Drs. Paul S. Albert (NIH Biostatistics) and Doron Levy (UMD Applied Mathematics)*
Co-advisors: *Drs. Jianxin Shi (NIH Biostatistics) and Hyokyoung G. Hong (NIH Biostatistics)*

2014/09 - 2016/06 **M.S. in Mathematics, University of Science and Technology of China** (USTC)
Dissertation title: *Ekeland Index Theory for Periodic Solutions and Symmetric Periodic Solutions of Convex Hamiltonian Systems*
Advisor: *Dr. Xinan Ma*
Co-advisor: *Dr. Hui Liu*

2010/09 - 2014/06 **B.S. in Mathematics and Applied Mathematics (Teacher Education Program), Nanjing Normal University** (NNU)
Dissertation title: *Some Problems in Hamiltonian Systems* (NNU Outstanding Graduation Thesis)
Advisor: *Dr. Yujun Dong*

2009/09 - 2010/06 **Undergraduate in Computer Science, NNU**

Employment History

2025/05 - present **Postdoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi)**
Biostatistics Branch (BB), Division of Cancer Epidemiology & Genetics (DCEG)
National Cancer Institute (NCI), National Institutes of Health (NIH)

2021/09 - 2025/05 **Predoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi, and Hyokyoung G. Hong)**
BB, DCEG
NCI, NIH

2019/08 - 2021/08 **Teaching Assistant**
Department of Mathematics, UMD

2016/08 - 2017/04 **Mathematics Teacher**
NO.1 Middle School Affiliated to Central China Normal University, Wuhan, China

Employment History (continued)

2014/09 - 2016/06 **Teaching Assistant**
Department of Mathematics, USTC

Research Interests

- **Longitudinal and correlated data methodology**, with emphasis on repeated-measures analysis, biomarker studies, and complex dependent data structures.
- **Statistical and computational methods in genetics and genomics** for analyzing electronic health records (EHR), real-world data (RWD), and large-scale population studies (e.g., the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer Screening Trial), with applications to prostate cancer risk stratification, early detection, and prediction.
- **Statistical and machine learning approaches in epidemiology**, including machine learning and deep learning for cancer genetic epidemiology, nutritional epidemiology, and metabolomics.

Publications

[*] corresponding author [†] co-first / equal contribution [‡] alphabetical order

Methodological / Statistical Papers

1. **Zhang, Pei**, X. Wang, Shi, Jianxin[*], and Albert, Paul S.[*], "Estimating the heritability of longitudinal rate-of-change: Genetic insights into PSA velocity in prostate cancer-free individuals," *Biostatistics*, 2026, *Biostatistics* (Accepted), preprint on arXiv [arXiv].
2. **Zhang, Pei**, Albert, Paul S.[*], and Hong, Hyokyoung G.[*], "Mixed modeling approach for characterizing the genetic effects in a longitudinal phenotype," *Annals of Applied Statistics*, vol. 19, no. 3, pp. 2070–2087, 2025, [PDF].

Collaborative / Applied Papers

1. H. Fan, H. Zhao, **Zhang, Pei**, P. Yu, Y. Ji, G. Chen, H. Jin, Y. Liu, J. Liu, Z.-S. Chen, Lyu, Aiping[*], Liang, Xinmiao[*], and Chen, Yang[*], "Homoisoflavanone delays colorectal cancer progression via DNA damage-induced mitochondrial apoptosis and parthanatos-like cell death," *Advanced Science*, e11406, 2026, [PDF].
2. H. Fan, H. Zhao, L. Gao, Y. Dong, **Zhang, Pei**, P. Yu, Y. Ji, Z.-S. Chen, X. Liang, and Chen, Yang[*], "CCN1 enhances tumor immunosuppression through collagen-mediated chemokine secretion in pancreatic cancer," *Advanced Science*, e2500589, 2025, [PDF].
3. Loftfield, Erikka[†*], **Zhang, Pei**[†], C. P. O'Connell, L. L. Kahle, K. Herrick, L. Abar, N. Khandpur, E. M. Steele, and H. G. Hong, "Performance of a food frequency questionnaire for estimating ultraprocessed food intake according to the NOVA classification system in the United States NIH-AARP diet and health study," *Journal of Nutrition*, vol. 155, no. 7, pp. 2376–2384, 2025, [PDF].
4. Wu, Yetian[†], **Zhang, Pei**[†], H. Fan, C. Zhang, P. Yu, X. Liang, and Chen, Yang[*], "GPR35 acts a dual role and therapeutic target in inflammation," *Frontiers in Immunology*, vol. 14, no. 1254446, 2023, [PDF].

Preprints / Under Review

1. Lim, Jungeun*, **Zhang, Pei**, J. Rhee, M. C. Playdon, A. J. Cross, R. Stolzenberg-Solomon, V. L. Roger, M. P. Purdue, Albanes, Demetrius*, Hong, Hyokyoung G.*, and Wong, Jason Y. Y.*, "Metabolomic profile score for predicting serum per- and polyfluoroalkyl substance (PFAS) levels: A novel application of machine learning to enhance detection of environmental pollutants," Under review at *International Journal of Hygiene and Environmental Health*, 2026.

PhD Thesis

1. **Zhang, Pei**, "Mixed modeling approaches for characterizing genetic effects and heritability metrics in longitudinal phenotypes," [PDF], Ph.D. dissertation, University of Maryland, College Park, 2025.

Honors

- 2026 — NCI Early K99/R00 Grant Application — Rated as "Competitive, Not Discussed"
- National Science Foundation (NSF) Travel Support Award, USA
- Institute of Mathematical Statistics (IMS) New Researcher Travel Award, USA
- 2025 & 2026 — NIH Postdoctoral Fellowship (twice)
- 2025 — NCI-DCEG Institutional Nominee for NCI Pathway to Independence Award for Outstanding Early Stage Postdoctoral Researchers (K99/R00) [about "Early K99" Grant - NIH] | [about "Early K99" Grant - Stanford Medicine] | [Notice of Funding Opportunity Number: PAR-23-286]
- NIH Fellows Award for Research Excellence (FARE) 2026 (top 25% of all applicants) [List of FARE 2026 Awardees Abstracts] | [about FARE]
- NIH Outstanding Graduate Research
- 2023 & 2024 — Women in Statistics and Data Science Conference Student Travel Award (twice)
- 2023 — UMD Jacob K. Goldhaber Travel Grant
- UMD International Conference Student Support Award
- 2022 & 2023 — UMD Graduate Student Travel Fellowship (twice)
- 2022 — Rutgers University Travel Award for Conference on Advances in Bayesian and Frequentist Statistics
- 2021 - 2024 — UMD Tuition Award for External Fellowship (4 times)
- NIH Predoctoral Fellowship (4 times) [about NIH Graduate Partnerships Program (GPP)]
- 2019 & 2020 — UMD Dean's Fellowship (twice)
- 2017 — First Award in Thesis Contest on Deep Integration of Key Competence and Classroom Teaching (NO.1 Middle School Affiliated to Central China Normal University)
- 2016 — USTC Outstanding Graduate
- 2014 & 2015 — USTC First Award of Academic Scholarship (twice)
- 2014 — NNU Outstanding Graduate
- NNU First Award of Feng Ruer Scholarship
- NNU Outstanding Dissertation
- 2013 — Certificate of Recommendation for Exemption from Entrance Examination—USTC Graduate Program
- Eligibility for Recommended Admission (top 5% in Mathematics Teacher Education Program, NNU)
- Second Award of Jiangsu Teaching Professional Skills Contest (ranked 4th in Mathematics across the Jiangsu Province)
- 2012 — NNU Second Award in Mathematical Modeling Contest

Honors (continued)

- NNU Outstanding Debate Team Award, Debate Competition on "Is Beauty Subjective or Objective?" Member of the award-winning team arguing that beauty is objective.

2010 & 2011
- NNU First Award of Rolling Scholarship (twice)

2009
- NNU First Award in Computer Contest

Academic Activities

Invited Talks / Seminars

2026
- International Biometric Conference (IBC), Seoul, South Korea
- Biostatistics Branch of NCI-DCEG Works in Progress, Rockville, Maryland, USA

Invited Posters / Panel Invitations

2026
- STATGEN 2026: Conference on Statistics in Genomics and Genetics, Emory University, Atlanta, Georgia, USA (Invited Poster)

2024
- Women in Statistics and Data Science (WSDS), Reston, Virginia, USA (Invited Panelist)

Contributed Presentations

2026
- The American Society of Human Genetics (ASHG) Annual Meeting, Montreal, Canada
- Joint Statistical Meetings (JSM), Boston, Massachusetts, USA
- The 11th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, Georgia, USA (Contributed Poster as a recipient of the IMS New Researchers Travel Award and NSF Travel Support Award)

2025
- Joint Statistical Meetings (JSM), Nashville, Tennessee, USA
- The 17th Annual NCI-DCEG Fellows' Training Symposium, Rockville, Maryland, USA

2024
- Joint Statistical Meetings (JSM), Portland, Oregon, USA
- AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA
- Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA
- NIH Annual Graduate Student Research Symposium, Bethesda, Maryland, USA

2023
- Women in Statistics and Data Science (WSDS), Seattle, Washington, USA
- Joint Statistical Meetings (JSM), Toronto, Canada
- Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA

2022
- Conference on Advances in Bayesian and Frequentist Statistics with a Celebration of the 80th Birthday of Professor William E. Strawderman, Rutgers University, New Brunswick, New Jersey, USA
- Annual Conference on Quantitative Approaches in Biology, Northwestern University, Chicago, Ohio, USA

Attendant

2026
- Artificial Intelligence (AI) Expo, The Department of Health and Human Services (HHS), Washington D.C., USA

Academic Activities (continued)

- NIH Library Artificial Intelligence (AI) Trainings led by OpenAI experts, Bethesda, Maryland, USA
- Biostatistics Symposium of South California (BSSC), Newport Beach, California, USA

2025 - Eastern North American Region (ENAR), New Orleans, Louisiana, USA

2024 - Statistics in the Age of AI Conference, The George Washington University (GWU), Washington DC, USA
- Eastern North American Region (ENAR), Baltimore, Maryland, USA

2023 - National Heart, Lung, and Blood Institute (NHLBI) Biostatistics Workshop on Clinical Trial Designs: Innovative Endpoints and Futility Monitoring, Bethesda, Maryland, USA

2022 - International Small Area Estimation (SAE) Annual Conference, UMD, College Park, Maryland, USA

2016 - National Tianyuan Foundation Summer School on Pure Mathematics, Summer Seminar on Pure Mathematics, Zhejiang University, Hangzhou, China

2013 - Yangtze River Delta Symposium on Partial Differential Equations and Doctoral Student Forum, Tongji University, Shanghai, China

Grant Writing Workshop

2026 - NCI K Grant Writing Workshop, Bethesda, Maryland, USA
- NCI-DCEG Writing Your Specific Aims Session, Rockville, Maryland, USA

2025 - NCI-DCEG Grant Writing Workshop, Rockville, Maryland, USA

Software

R Packages and Research Code

- **Heritability-Velocity** — Author and Maintainer. Reproducible framework for joint estimation of static and dynamic heritability components in longitudinal phenotypes, featuring scalable AI-REML and REHE algorithms for genome-wide data. [GitHub] | [arXiv]

- **PLCO** — Author and Maintainer. Mixed-effects modeling framework with crossed genetic and individual-specific random effects to quantify genetic influences on PSA trajectories in large-scale prostate cancer screening studies. [GitHub] | [Paper]

- **AARP** — Author and Maintainer. Implementation of measurement error models for bias correction in food frequency questionnaire–based dietary intake assessment. [GitHub] | [Paper]

Programming and Statistical Computing

- R | Python | C/C++ | SQL | Jupyter Notebook

Research Infrastructure

- NIH Biowulf High-Performance Computing Cluster | All of Us Research Hub | UK Biobank

Scientific Writing and Reproducibility

- LaTeX | Microsoft Word | R Markdown | Git

Teaching

Teaching Assistant / Grader

Fall 2019 — **Sampling Theory (STAT440)**, Undergraduate/master-level.
Grader, UMD

Spring 2020 — **Introduction to Linear Algebra (MATH240)**, Undergraduate-level.
Teaching Assistant, UMD

Summer 2020 — **Introduction to Mathematical Proof (MATH310)**, Undergraduate-level.
Grader, UMD

Introduction to Probability Theory (STAT410), Undergraduate/master-level.
Grader, UMD

Fall 2020 & Spring 2021 — **Elementary Probability and Statistics (STAT100)**, Undergraduate-level.
Teaching Assistant, UMD

Spring 2015 & 2016 — **Calculus II**, Undergraduate-level.
Teaching Assistant, USTC

Fall 2016 — **Calculus I**, Undergraduate-level.
Teaching Assistant, USTC

Mathematics Teacher (Internship)

2013/09 - 2013/11 — **High School Mathematics**. High school level.
Mathematics Teacher Intern, Jinling High School, Nanjing, China

Service

Journal Reviewer

2021 - present — Biometrics (1); Statistics in Medicine (2); Journal of Biopharmaceutical Statistics (1); Frontiers in Oncology (1); LabMed Discovery (1)

Judging

2026 — Chief Judge, NIH Fellows Award for Research Excellence (FARE) 2027 Competition

Conference / Seminar Organizer

2024 — Co-organizer of the panel "Discussion of Mentoring Program and Participant Experiences" at WSDS 2024

Chair of the session "Rethinking Current Modeling Strategies: Advances and Applications" at JSM 2024

2023 — Chair of the session "Concurrent - Statistical Insights into Mortality and Inference: From Choquet Capacities to Demographic Patterns" at WSDS 2023

Mentoring

2025–present — Co-mentor (with **Paul S. Albert, PhD**) of postbaccalaureate fellow **Sumaja Bandreddi** (M.S., B.A., University of Virginia), Biostatistics Branch, DCEG, NCI, NIH

Spring 2021 — Directed Reading Program mentor for undergraduate **Andrew Craun**, Department of Mathematics, University of Maryland, College Park

Service (continued)

Leadership

- 2026 — Member, NIH Fellows Award for Research Excellence (FARE) 2027 Committee
- 2022–2023 — Vice President, Women in Mathematics, University of Maryland, College Park
- 2020–2021 — Representative, Graduate Student Government, University of Maryland, College Park
- 2010–2011 — Team Leader, Volunteer Interpreters Group, Nanjing Yunjin Museum, Nanjing, China

Volunteer Service

- 2026 — Volunteer, The Chinese Biopharmaceutical Association-USA (CBA)
- 2025 — Volunteer, Sino-American Pharmaceutical Professionals Association (SAPA)-DC
- 2022 — Volunteer Interviewee, The Center for Mathematics Education, University of Maryland, College Park
- 2009–2010 — Volunteer Teacher, Nanjing Hongshan Primary School for Children of Migrant Workers, Nanjing, China

Memberships

- 2026 - present — American Society of Human Genetics
- 2024 - present — Washington Statistical Society
  - International Chinese Statistical Association
  - Section of Statistics in Genetics and Genomics
- 2023 - present — American Statistical Association
  - Institute of Mathematical Statistics
  - Society for Mathematical Biology
  - Caucus for Women in Statistics and Data Science
  - Eastern North America Region/International Biometric Society
- 2019 - 2025 — American Mathematical Society

References

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Jianxin Shi
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Doron Levy
Professor of Mathematics
Chair, Department of Mathematics
Director, Brin Mathematics Research Center
Co-director, the NCI-UMD Partnership for Integrative Cancer Research
Fellow of the Society for Industrial and Applied Mathematics (SIAM)
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